A Longitudinal MEG-EEG Based System Design for Parkinson's Disease Patient Segmentation and Progression Analysis

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Abstract

Parkinson's Disease (PD) functions as a progressive neurological disorder which brings motor control impairments and cognitive function deficits. Patient groups become difficult to establish because patients display different disease patterns which make it hard to measure disease progression during their lifetime. This report describes a system which performs analysis on combined magnetoencephalography (MEG) and electroencephalography (EEG) data collected across multiple time periods. The process includes preprocessing, feature extraction, feature integration, similarity analysis, and classification. The system groups patients into three categories representing different stages of disease progression based on neural similarity patterns. This structured and objective approach supports long-term monitoring of Parkinson's Disease.

1 Introduction

Parkinson's Disease (PD) advanced neurodegenerative disorder which primarily disrupts motor functions. The condition causes patients to develop symptoms which include tremors and rigidity and bradykinesia and postural instability. The symptoms emerge from the progressive loss of dopamine-producing neurons which leads to observable brain activity shifts throughout time.

The Unified Parkinson's Disease Rating Scale UPDRS functions as a clinical assessment tool which medical professionals use to determine the severity of Parkinson's disease. The assessments provide only temporary evaluations because they do not detect minor changes in brain activity that occur throughout multiple patient appointments. Thus researchers require development of objective systems which use signal detection methods to track disease advancement.

Non-invasive neuroimaging techniques operate through magnetoencephalography

MEG and electroencephalography EEG to capture brain activity with precise timing abilities. MEG delivers better spatial precision whereas EEG records rapid alterations in brain activity. The two methods together provide researchers with comprehensive methods to study brain activity patterns and their progression across different time periods.

This project proposes a longitudinal MEG-EEG based system to analyze neural signals, extract meaningful features, and group patients based on similarity in brain activity patterns. The researchers conduct experimental validation of their work with the EEG Motor Movement/Imagery dataset from PhysioNet because no public datasets for longitudinal Parkinson's research exist. The dataset functions as a benchmark to assess preprocessing methods and feature extraction techniques and classification systems and analysis methods that apply to neurological signal processing.

The assessment of the regression model and the cluster model is done by using the extracted features of the EEG, which the study applies to the model behavior assessment, apart from the classification analysis. The study applies Linear Regression to determine the relationship that exists among the different features of the EEG, whereas the K-Means cluster model is used to determine the natural grouping of the data. The additional experiment offers more knowledge regarding the operation of the features with the performance capabilities of the model.

2 Literature Survey

The progression of Parkinson's Disease has been studied through multiple research studies which showed that Parkinson's Disease develops through alterations of brain rhythms and functional connectivity and synchronization patterns which scientists detected by using EEG and MEG. The disease progresses based on longitudinal studies which show that beta-band activity declines and motor cortex activation changes and connectivity decreases throughout the disease process.

Previous research shows that using multiple EEG and MEG features which include spectral and statistical and connectivity metrics leads to better classification results than using a single feature method. The application of machine learning techniques which include k-Nearest Neighbor (kNN) and Support Vector Machines (SVM) and Random Forests and Neural Networks has reached widespread use for EEG-based classification tasks which mainly focus on motor imagery and movement analysis.

Studies which examine EEG motor imagery classification have found that frequency-domain and statistical features enable researchers to distinguish between left-hand and right-hand motor tasks. The existing research studies focus mainly on achieving high classification accuracy yet they fail to investigate how models function because they do not understand overfitting and underfitting and class separability.

The current research advances existing literature through its use of an easy-to-understand feature-based method which employs kNN classification. The academic

research environment benefits from this method because it needs small datasets for testing and it allows researchers to see how different classifier parameters especially k value impact performance and generalization and model stability.

3 Methodology

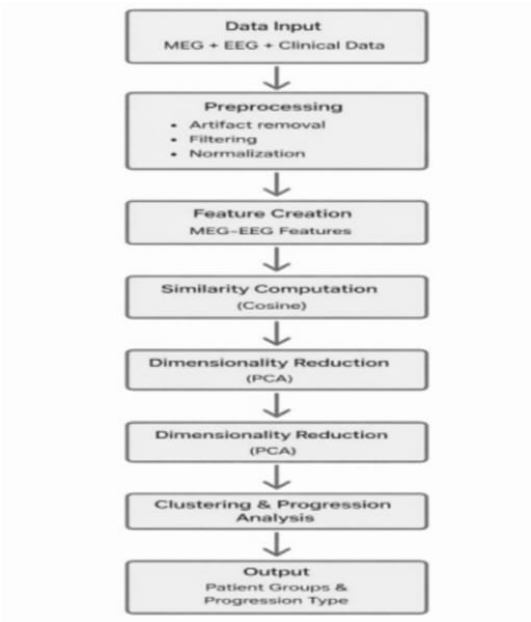

Figure 1: Block Diagram of the Proposed MEG-EEG System

3.1 System Overview

The proposed system is designed to analyze neurological EEG signals and evaluate classification performance through experimental validation. The framework aims to provide longitudinal support for Parkinson's Disease research through its integration of MEG and EEG data but currently works to validate its signal processing and classification pipeline through tests on a public EEG motor imagery dataset. The system follows a structured sequence of operations which includes data ingestion, preprocessing, feature extraction, similarity analysis and patient segmentation. The systematic approach of the study establishes a method that maintains both consistency and reliability for the examination of neural activity patterns.

3.2 Data Ingestion

The system begins with the ingestion of EEG data from the EEG Motor Movement/Imagery dataset available on PhysioNet. The dataset consists of multichannel EEG recordings that were collected during left-hand and right-hand motor imagery experiments. The recording provides task labels which let supervised learning proceed. The dataset functions as a reference point for testing the proposed methodology while demonstrating how classifiers react to brain signal data because longitudinal Parkinson's Disease datasets are difficult to obtain for educational research purposes.

3.3 Preprocessing

EEG recordings often contain unwanted noise through muscle activity and eye movements together with electrode displacement and environmental interference. The process of preprocessing must be conducted because it brings better signal quality and keeps all trial results consistent. The process applies noise reduction techniques to remove artifacts while conducting signal normalization which establishes consistent EEG signal standards. The preprocessing method establishes better feature extraction reliability which leads to enhanced clas-

sification results.

3.4 Feature Extraction and Validation

The process of EEG signal analysis starts with preprocessing which extracts vital features that show brain activity. Researchers calculate frequency-domain features by measuring signal power in standard EEG bands which include delta, theta, alpha, beta, and gamma bands. The frequency bands serve as important links to both motor control activities and cognitive functioning. The process extracts both spectral and basic statistical features which include mean, variance, and standard deviation to measure the complete signal distribution and its changes.

The process of feature validation evaluates which extracted features hold importance and which features provide duplicate information. The process maintains essential informative features which help the classification model to function with precise and reduced feature sets.

3.5 Similarity Computation

The process of similarity computation establishes relationships among the extracted feature vectors. The system employs cosine similarity together with Euclidean distance to measure similarity between continuous-valued features. The measures establish the level of connection between different EEG signal samples through their extracted features. Researchers employ similarity analysis to measure motor imagery class separation which enables them to proceed with segmentation and classification processes.

3.6 Patient Segmentation

Patients are grouped into clusters based on similarity scores. Each cluster represents patients with similar neural activity patterns and can be interpreted as:

- Stable patients
- Slow-progressing patients
- Fast-progressing patients

  The system uses similarity scores together with classification results to organize EEG signal samples into specific categories that show matching neural activity patterns. The segmentation step demonstrates how neural patterns can be grouped based on feature similarity even though the experimental dataset focuses on motor imagery tasks instead of clinical Parkinson's data. Researchers use Parkinson's Disease segmentation to create three patient categories which include stable patients slow-progressing patients and fast-progressing patients through assessment of neural activity patterns across time.

3.7 Experimental Setup

The proposed system undergoes experimental testing through the EEG Motor Movement/Imagery dataset which exists as an accessible resource at PhysioNet. The dataset contains multiple-channel EEG data which researchers collected during various motor imagery experiments. The study focuses on classifying EEG signals which match left-hand and right-hand motor imagination activities.

After the preprocessing stage and feature extraction process the researchers created two distinct dataset partitions which included a training set and a testing set that used a 70:30 split ratio. The researchers employed a k-Nearest Neighbor (kNN) classifier to conduct classification experiments which involved testing different k values to determine their impact on system performance. The researchers used accuracy, precision, and recall as standard evaluation metrics to measure how well the classifier performed.

3.8 Linear Regression Analysis

The researchers employed a Linear Regression model to train the selected EEG band power features to study the relationship between individual features and the target values. The analysis was performed using both single-feature and multi-feature regression models to evaluate their impact on prediction performance. The dataset was divided into training and testing sets using a consistent split ratio. This experiment evaluated the model's ability to generalize and its prediction behavior across datasets.

3.9 Regression Evaluation Metrics

The regression model performance was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the R^2 score. These metrics were computed for both training and testing datasets to assess prediction accuracy and to identify overfitting or underfitting behavior.

3.10 K-Means Clustering

K-Means clustering was performed on the extracted EEG features after removing all non-numeric attributes. Initially, clustering was carried out with two clusters (k = 2). The clustering quality was evaluated using the Silhouette Score, Calinski-Harabasz Score, and Davies-Bouldin Index. Further experiments were conducted by varying the value of k from 2 to 10 to identify the optimal number of clusters. An elbow plot based on inertia values was generated to support optimal cluster selection.

3.11 Decision Tree Based Analysis

For the experiment, the entropy value is calculated to find the impurities of the EEG data set. The Gini index value is also calculated for finding the impurities of the target class data set. The information gain value is also calculated for each feature to find the most important feature for the root node of the Decision Tree Classifier.

As there are many features in the EEG data set with continuous values, equal width binning is applied for converting the values into categorical values. This is necessary for calculating the entropy value and information gain value.

After finding the important features for the Decision Tree Classifier, the Decision Tree Classifier is implemented based on the entropy values. The Decision Tree Classifier is implemented for the EEG data set feature set with the target variable as the

subject label. The tree structure is also visualized based on the Decision Tree Classifier. In addition, two features are also selected from the EEG data set for visualizing the decision boundary of the Decision Tree Classifier.

4 Results Analysis and Discussion

The EEG Motor Movement/Imagery dataset delivered experimental results that showed the extracted features successfully distinguished between left-hand and right-hand motor imagery tasks. EEG signals require study because they produce natural noise which leads to overlapping patterns. The frequency band together with statistical features successfully detects motor activity. The selected features enable classification because they deliver sufficient class separation.

The k-Nearest Neighbor (kNN) classifier demonstrates different performance results through its various k value settings. The classifier reaches high training dataset accuracy results through its use of k values. The testing accuracy results show lower performance under these conditions. The classifier develops excessive sensitivity towards specific data points plus EEG signal noise which causes it to overfit. The model learns training data through memorization instead of developing generalizable patterns which results in reduced performance on new data.

The classifier uses a larger neighborhood to make predictions when the value of k increases. This process produces smoother decision boundaries which enhances generalization. The testing accuracy difference from training accuracy decreases at moderate k values which shows the model fits regular patterns. The classifier achieves effective generalization through this condition because it reaches a balance between bias and variance while preserving key class-specific information.

The performance of the classifier decreases on both training and testing databases when k values exceed their optimal range. The behavior shows that the system suffers from underfitting because the classifier develops excessive generalization which stops it from detecting essential motor imagery class differences. The model fails to recognize vital local EEG data patterns because excessively high k values diminish the impact of nearby relevant samples.

The kNN classifier shows good performance results when the correct k value is chosen according to the complete evaluation which includes accuracy and precision and recall measurement. The testing and training results demonstrate how parameter adjustment leads to maximum performance achievement. The observed trends in overfitting, regular fitting, and underfitting clearly demonstrate how classifier behavior is influenced by model parameters.

The results demonstrate that the kNN classifier functions correctly for EEG-based motor imagery classification while also showing how the model operates. The current experiments use motor imagery EEG data but the results show that the proposed framework can analyze Parkinson's Disease progression through actual clinical datasets which will be available in the future.

4.1 Linear Regression Results

The linear regression experiments confirmed the presence of relationships among the EEG features. The evaluation metrics obtained during training and testing demonstrate the behavior of the regression model. When multiple features were included, the regression model achieved a higher R^2 score, indicating improved explanatory power and the influence of multiple EEG attributes on prediction performance.

4.2 Clustering Results

K-Means clustering was performed with an initial cluster count of $k = 2$. The Silhouette Score reflected the degree of separation between clusters, while the Calinski-Harabasz Score indicated cluster compactness. The Davies-Bouldin Index measured cluster similarity. By varying the value of k between 2 and 10, the clustering performance showed notable variations, with the optimal cluster count corresponding to the maximum Silhouette Score.

4.3 Elbow Plot Interpretation

The elbow plot illustrates the relationship between inertia and the number of clusters. As the number of clusters increases, inertia decreases due to reduced intra-cluster variance. Beyond a specific cluster count, the reduction in inertia becomes gradual, indicating the elbow point and suggesting an optimal number of clusters for K-Means clustering.

4.4 Decision Tree Experiment Results

The calculation of entropy and Gini index gave us an idea of the impurity level of the data set. The Information Gain calculation gave us an idea of the most important feature that would be used as the root node for the Decision Tree.

The visualization of the Decision Tree gave us an idea of how the data set is being recursively split based on the values of the EEG features for classification.

The visualization of the decision boundary for two EEG features gave us an idea of how the classifier would split the feature space for classification.

From these results, we can see that EEG features have information that can be used for classification, and the Decision Tree model is capable of learning from it.

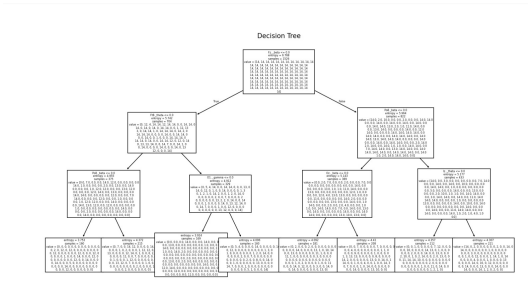

Figure 2: Decision Tree constructed using EEG features

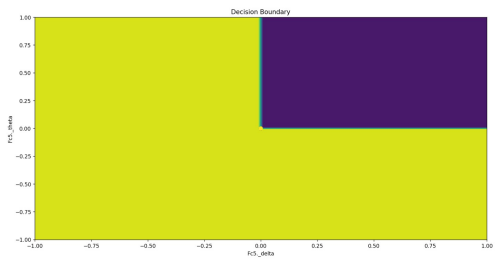

Figure 3: Decision boundary visualization using two EEG features

5 Conclusion

The researchers carried out regression analysis and clustering experiments in addition to classification to examine the behavior of EEG-based models on extracted features. The regression analysis revealed meaningful relationships among EEG band power features, while clustering experiments identified natural groupings within the dataset. Performance evaluation metrics demonstrated different model states, including overfitting, regular fitting, and underfitting.

The Decision Tree experiment provided more information about the importance of EEG features with entropy and information gain. The visualization of the decision tree and the decision boundary was also useful in understanding the workings of the classifier for the given classes based on the features.

These experiments enhanced the robustness of the proposed framework and establish a foundation for future validation using real Parkinson's Disease clinical datasets.

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